Technical Supplement:
When Short Trajectories Lie – IIGC Framework Details and Extended Results

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Environment Details

510K Card Game

510K is a four-player trick-taking game with a standard 52-card deck. Cards 5, 10, K carry point values (5, 10, 10). The lead player sets the pattern (single, pair, straight, bomb, 510K pattern, etc.); others must follow with a higher-rank play or pass. The trick winner collects all 5/10/K points from the trick.

Action space: 300 discrete actions (0 = pass, 1–299 = pattern index). **Observation space:** 112-dim Box (116-dim for OBVIOUS mode, with 4-bit teammate indicator). **15 pattern types:** SINGLE, PAIR, THREE, THREE_ONE, THREE_PAIR, STRAIGHT (5+), CONSECUTIVE_PAIRS (3+), AIRPLANE_NONE, AIRPLANE_SINGLE, AIRPLANE_PAIR, BOMB, NONSUIT_510K, SUIT_510K, RED_A_SINGLE, RED_A_PAIR.

Four modes share identical card mechanics but differ in team structure:

- SINGLE: solo-competitive. Game ends when first player empties hand. Winner $+30$, losers -10 each (plus own 510K points).
- STATIC: fixed teams (players 0&2 vs. 1&3). Both teammates must finish. Winners $\times 2$ multiplier, losers $\times 1$.
- DYNAMIC: teams determined by hidden red-Ace distribution (holders of A♡/A♢ form one team). Same scoring as STATIC. Red A is the strongest single; red-A pair is the strongest bomb.
- OBVIOUS: identical card mechanics to DYNAMIC, but team membership is observable as a 4-bit observation. This isolates *information structure* from card mechanics.

Toy Matching Environment

A minimal 2-action contextual bandit that isolates IIGC:

- **State**: 1-dim (constant 0) in HIDDEN; 2-dim (one-hot partner type) in REVEALED.
- **Action**: $a \in \{0, 1\}$. Agent chooses which partner type to match.
- **Partner**: hidden binary type $p \in \{0, 1\}$, fixed per episode.
- **Reward**: $+1$ if $a = p$, -1 if $a \neq p$.
- **Training**: PPO, 10K steps, 10 seeds per condition.

In HIDDEN mode, the agent receives no information about p. Gradients from episodes where $p = 0$ and $p = 1$ cancel completely ($\kappa \approx 0$). In REVEALED mode, the agent observes p directly and learns the optimal type-matching policy.

Overcooked Environment

Grid-world cooperative cooking game with configurable partner roles:

- **State**: 96-dim observation (grid layout, object positions, inventory).
- **Action**: 6 discrete actions (up/down/left/right/interact/stay).
- **Partner roles**: CHEF (cooks soup) or WAITER (delivers soup).
- **Static mode**: partner role visible as 1-hot observation.
- **Dynamic mode**: partner role hidden, switches every 30 steps.
- **Reward**: sparse $+20$ per soup delivered.
- **Training**: PPO, 1M steps, 8 parallel envs, 8 seeds per mode.

κ Computation

Pseudocode

Algorithm 1 Computing the Relational Gradient Retention Ratio κ

Input: Trained policy π_θ, environment with two hidden configurations A and B, number of rollouts N, discount γ
Output: $\kappa \in [0, 1]$

$g_A \leftarrow \mathbf{0}$ ▷ Accumulated gradient for config A
$g_B \leftarrow \mathbf{0}$ ▷ Accumulated gradient for config B
for $i \leftarrow 1$ to N **do**
  $\tau_A \leftarrow \text{Rollout}(\pi_\theta, \text{config} = A)$
  $\tau_B \leftarrow \text{Rollout}(\pi_\theta, \text{config} = B)$
  $g_A \leftarrow g_A + \frac{1}{N} \sum_t \nabla_\theta \log \pi_\theta(a_t|o_t) \cdot R(\tau_A)$
  $g_B \leftarrow g_B + \frac{1}{N} \sum_t \nabla_\theta \log \pi_\theta(a_t|o_t) \cdot R(\tau_B)$
end for

$g_{\text{mixed}} \leftarrow \frac{1}{2}(g_A + g_B)$
$\kappa \leftarrow \dfrac{\|g_{\text{mixed}}\|^2}{\frac{1}{2}(\|g_A\|^2 + \|g_B\|^2)}$
return κ

Implementation Notes

- Gradients are computed via REINFORCE $\nabla_\theta \log \pi(a|o) \cdot R(\tau)$, not via backpropagation through a training loss.
- κ requires only **rollout access** to the policy, not training access. It can be computed post-hoc from any saved checkpoint.
- $N \approx 30$ rollouts per configuration suffices for stable κ estimates.
- For value-based methods (DQN), we substitute the TD-loss gradient $\nabla_\theta (Q_\theta(s, a) - y)^2$ in place of the REINFORCE gradient.
- When $\|g_A\|^2 + \|g_B\|^2 = 0$ (zero gradient energy), define $\kappa = 0.5$.

Proofs of Propositions

Proposition 1 (Pythagorean Decomposition)

Let $g_A, g_B \in \mathbb{R}^d$ be the policy gradients under configurations A and B. Define $g_+ = \frac{g_A + g_B}{2}$ (shared component) and $g_- = \frac{g_A - g_B}{2}$ (differential component).

$$\|g_A\|^2 + \|g_B\|^2 = \|g_+ + g_-\|^2 + \|g_+ - g_-\|^2 \tag{1}$$
$$= (\|g_+\|^2 + \|g_-\|^2 + 2g_+ \cdot g_-) + (\|g_+\|^2 + \|g_-\|^2 - 2g_+ \cdot g_-) \tag{2}$$
$$= 2(\|g_+\|^2 + \|g_-\|^2) \tag{3}$$

Therefore $\frac{\|g_A\|^2 + \|g_B\|^2}{2} = \|g_+\|^2 + \|g_-\|^2$. Since $\kappa = \|g_+\|^2 / (\|g_+\|^2 + \|g_-\|^2)$, it follows that κ is the fraction of total gradient energy retained after aggregation.

Proposition 2 (Effective Learning Signal)

Under balanced configuration mixing ($p_A = p_B = 1/2$), the expected per-step gradient during training is:

$$\mathbb{E}[g_{\text{train}}] = \frac{1}{2}(g_A + g_B) = g_+ \tag{4}$$

The effective gradient magnitude satisfies:

$$\|g_+\|^2 = \kappa \cdot \frac{\|g_A\|^2 + \|g_B\|^2}{2} \tag{5}$$

When $\kappa \to 0$, $\|g_+\|^2 \to 0$ and the policy receives no effective learning signal regardless of the per-configuration gradient magnitudes.

Proposition 3 (Gradient Structure Distinction)

For policy-gradient methods, g_r^{PG} reflects the direction of improvement in action preferences for configuration r. When optimal actions differ across configurations, $g_A^{\text{PG}} \approx -g_B^{\text{PG}}$ and $\kappa_{\text{PG}} \to 0$.

For TD-learning methods, g_r^{TD} minimizes prediction error under a *shared* Q-function. Both configurations push toward representations that minimize error across the mixture, producing aligned updates: $\kappa_{\text{TD}} > \kappa_{\text{PG}}$ under identical information asymmetry.

Full Path Integral Data

Policy (seed)	Path Length	Endpoint Dist.	Curvature
SINGLE (41)	0.340	0.046	7.3×
SINGLE (51)	0.482	0.068	7.1×
SINGLE (52)	0.561	0.034	16.6×
SINGLE (53)	0.454	0.108	4.2×
SINGLE (54)	0.443	0.051	8.8×
STATIC (41)	0.253	0.071	3.6×
STATIC (61)	0.342	0.086	4.0×
STATIC (62)	0.490	0.073	6.7×
STATIC (63)	0.281	0.048	5.9×
STATIC (64)	0.278	0.021	13.5×
DYNAMIC (41)	0.299	0.046	6.5×
DYNAMIC (42)	0.231	0.112	2.1×
DYNAMIC (43)	0.367	0.026	14.3×
DYNAMIC (44)	0.274	0.031	8.9×
OBVIOUS (71)	0.203	0.100	2.0×
OBVIOUS (72)	0.355	0.122	2.9×
OBVIOUS (73)	0.366	0.051	7.2×
OBVIOUS (74)	0.302	0.022	13.6×
OBVIOUS (75)	0.318	0.036	8.9×
OBVIOUS (76)	0.391	0.085	4.6×
OBVIOUS (77)	0.283	0.043	6.6×
OBVIOUS (78)	0.404	0.035	11.5×

Table 1: Full path integral data for all 22 seeds across 4 modes.

Env	Algo	Mode	κ (mean $\pm$ std)	n
Toy	PPO	HIDDEN	0.039 ± 0.073	10
Toy	PPO	REVEALED	0.726 ± 0.100	10
Toy	A2C	HIDDEN	0.195 ± 0.352	10
Toy	A2C	REVEALED	0.842 ± 0.025	10
510K	A2C	SINGLE	0.644 ± 0.201	8
510K	A2C	STATIC	0.500 ± 0.000	8
510K	A2C	DYNAMIC	0.519 ± 0.060	8
510K	DQN	SINGLE	0.797 ± 0.123	8
510K	DQN	DYNAMIC	0.917 ± 0.064	8
510K	SAC	SINGLE	0.504 ± 0.038	2
510K	SAC	DYNAMIC	0.540 ± 0.069	2
510K	PPO	SINGLE	0.569	1
510K	PPO	DYNAMIC	0.444 ± 0.069	9
510K	REINFORCE	SINGLE	0.487 ± 0.312	8
510K	REINFORCE	DYNAMIC	0.605 ± 0.238	8
Overcooked	PPO	STATIC	0.500 ± 0.006	8
Overcooked	PPO	DYNAMIC	0.000 ± 0.000	8
Overcooked	A2C	STATIC	0.500 ± 0.000	8
Overcooked	A2C	DYNAMIC	0.125 ± 0.217	8
Overcooked	DQN	STATIC	0.473 ± 0.093	8
Overcooked	DQN	DYNAMIC	0.645 ± 0.209	8

Table 2: Complete κ dataset across all environments, algorithms, and modes.

Reveal Fraction	κ (mean $\pm$ std)	n
0%	0.534 ± 0.038	2
25%	0.446 ± 0.085	2
50%	0.514 ± 0.024	2
75%	0.324 ± 0.079	2
100%	0.616 ± 0.039	2

Table 3: Continuous reveal κ values at five information levels (510K, PPO).

Extended κ Data

Continuous Reveal Data

Training Hyperparameters

Computing Infrastructure

- GPU: NVIDIA GeForce RTX 5070 Ti Laptop GPU (12 GB VRAM)
- CPU: Intel(R) UHD Graphics (128 MB, auxiliary)
- RAM: 32 GB
- OS: Windows 11
- Framework: PyTorch 2.x, stable-baselines3 2.x, sb3-contrib 2.x
- Environment: Gymnasium 0.29+, NumPy 1.24+

Code Availability

All source code is publicly available under the MIT License:

- **Main repository** (full experimental pipeline): https://github.com/flavieninpekin/Information-Induced-Gradient-Contraction-in-Partially-Observable-MARL

Parameter	Value
Learning rate	3×10^{-4}
PPO n_steps	2048 (self-play) / 256 (parallel)
Batch size	64 / 256
Epochs per update	10
γ (discount)	0.99
GAE λ	0.95
Clip range	0.2
Entropy coeff.	0.01
Network	MLP [256, 256], tanh activation
A2C n_steps	256
DQN buffer size	50,000
DQN learning starts	5,000
DQN τ (target update)	0.005
DQN exploration	ϵ-greedy: $1.0 \rightarrow 0.02$ over 30% of training
SAC α	Auto-tuned (initial 0.2)
REINFORCE lr	1×10^{-3}

Table 4: Training hyperparameters for all algorithms.

- **Reproduction package** (curated for paper replication): https://github.com/flavieninpekin/AAAI2027-510k-clear
- **510K environment** (standalone Gymnasium + PettingZoo package): https://github.com/flavieninpekin/510k$_{e}nv$

The reproduction package includes pre-computed results for validation, config files for all experiment configurations, and a single-entry `experiments/reproduce_all.py` script to reproduce all paper results.